

```
In [ ]: #Conditions
```

```
In [1]: def studentstatus(marks):  
        if marks >= 70:  
            print("Student got 1st Class")  
        elif marks >= 50 and marks < 70:  
            print("Student got 2nd Class")  
        elif marks >=35 and marks < 50:  
            print("Student got 3rd Class")  
        else :  
            print("Student Fail")
```

```
In [2]: studentstatus(34)
```

Student Fail

```
In [4]: studentstatus(77)
```

Student got 1st Class

```
In [5]: studentstatus(56)
```

Student got 2nd Class

```
In [6]: studentstatus(44)
```

Student got 3rd Class

```
In [ ]: #Cricket example
```

```
In [10]: def playerscore_status(runs):  
        if runs >= 200:  
            print("Player done double century")  
        elif runs >= 100 and runs < 200:  
            print("Player done century ")  
        elif runs >=50 and runs < 100:  
            print ("Player done half century")  
        else:  
            print("Player not done any century")
```

```
In [11]: playerscore_status(240)
```

Player done double century

```
In [12]: playerscore_status(157)
```

Player done century

```
In [13]: playerscore_status(77)
```

Player done half century

```
In [14]: playerscore_status(44)
```

Player not done any century

```
In [ ]: #Mini Calculator
```

```
In [2]: def add(a,b):
        c=a+b
        print("The summation of two numbers is :",c)
        def sub(a,b):
            c=a-b
            print("The subtraction of two numbers is :",c)
        def mul(a,b):
            c=a*b
            print("The multiplication of two numbers is :",c)
        def div(a,b):
            c=a/b
            print("The division of two numbers is :",c)
```

```
In [7]: print("Enter 1st number")
        a = int(input())
        print('Enter Operation: add/sub/mul/div')
        operation = input()
        print("ENter 2nd number")
        b = int(input())
        if operation == "add":
            add(a,b)
        elif operation == "sub":
            sub(a,b)
        elif operation == "mul":
            mul(a,b)
        elif operation == "div":
            div(a,b)
        else:
            print("Enter Valid operator")
```

Enter 1st number

99

Enter Operation: add/sub/mul/div

dre

ENter 2nd number

34

Enter Valid operator

```
In [ ]: #Operators
```

```
In [ ]: #1. Arthamatic Operators
```

```
In [8]: a = 20  
b = 6
```

```
In [9]: print("sum :",a + b) #addition  
print("sub :",a - b) #subtraction  
print("mul :",a * b) #multiplication  
print("div1 :", a / b) #division1  
print("div2 :", a // b) #division2  
print("mod :", a % b) #mod  
print("power :", a ** b) #power
```

```
sum : 26  
sub : 14  
mul : 120  
div1 : 3.3333333333333335  
div2 : 3  
mod : 2  
power : 64000000
```

```
In [ ]: #2.Comparison / relational operators
```

```
In [10]: x = 7  
y = 5
```

```
In [11]: print( x == y)
```

```
False
```

```
In [12]: print ( x != y)
```

```
True
```

```
In [13]: print( x >= y)
```

```
True
```

```
In [14]: print ( x <= y)
```

```
False
```

```
In [ ]: #3. Logical operators
```

```
In [15]: r = 7
```

```
In [16]: print ( r > 5 and r < 9 ) # both sholud be true
```

True

```
In [17]: print( r > 5 or r == 9 ) #any one sholud be satisfied
```

True

```
In [ ]: #4. Bit wise operators
```

```
In [18]: print ( 5 & 3 )
```

1

```
In [19]: print( 5 | 3 )
```

7

```
In [ ]: #-----
```

```
In [10]: def employee_performance(rating):  
    if rating >= 8:  
        print("Excellent Performar")  
    elif rating >= 5 and rating < 8:  
        print("Good Performar")  
    elif rating >= 2 and rating < 5:  
        print("Average Performar")  
    else:  
        print("Poor Perfomar")
```

```
In [5]: employee_performance(8.4)
```

Excellent Performar

```
In [6]: employee_performance(5.8)
```

Good Performar

```
In [8]: employee_performance(3)
```

Average Performar

```
In [9]: employee_performance(1.5)
```

Poor Perfomrar

```
In [ ]: #mini calculator Logic
```

```
In [11]: def add(a,b):
          c = a + b
          print("The summation of two numbers is :",c)
def sub(a,b):
          c = a - b
          print("The subtraction of two numbers is :",c)
def mul(a,b):
          c = a * b
          print("The multiplication of two numers is :",c)
def div(a,b):
          c = a / b
          print("The division of two numbers is :",c)
def power(a,b):
          c = a ** b
          print("The power of two numbers is :",c)
```

```
In [25]: print("Enter 1st number")
          a = int(input())
          print("Enter operator: + | - | * | / | ** ")
          operator = input()
          print("Enter 2nd number")
          b = int(input())
          if operator == '+':
              add(a,b)
          elif operator == '-':
              sub(a,b)
          elif operator == '*':
              mul(a,b)
          elif operator == '/':
              div(a,b)
          elif operator == '**':
              power(a,b)
```

Enter 1st number

4

Enter operator: + | - | * | / | **

**

Enter 2nd number

5

The power of two numbers is : 1024

In []:

In []: